



MARRI LAXMAN REDDY INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(AN AUTONOMOUS INSTITUTION)

(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)

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Abstract Proforma

Field based Project/ Industry Oriented Mini Project/Major Project

Academic Year: 2025-2026		Regulation: MLRS-R22		
Branch : CSM	Year/Sem: 4-I	Section: B	Project Batch No.: B20	
Registration Details	S. No	Name of the student	Roll Number	Signature
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Name of the Guide & Designation		MR.K. SRIKANTH ASSISTANT PROFESSOR		
Area (Domain) of the Project		Artificial Intelligence and Multi-Agent Systems		
Title of the Project (Tentative)		OWL: Optimized Multi-Agent Workforce for Real-World Automation		
Type of the Project		Software application/ Public Service / Other		
Abstract				
• Background/Introduction: OWL (Optimized Workforce Learning) is an advanced AI-driven system designed to enhance workforce efficiency through multi-agent collaboration and real-world task automation. By integrating intelligent learning algorithms, OWL optimizes human-AI interactions, streamlines complex workflows, and enables adaptive decision-making, transforming traditional workforce management into a dynamic, data-driven ecosystem. • Objectives: The primary objectives of this project are to: 1. To develop an intelligent multi-agent system that enhances workforce collaboration and task automation in real-world environments. 2. To optimize learning and adaptability of agents through continuous data-driven feedback and performance analysis.				



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3. To improve overall productivity, decision-making efficiency, and operational accuracy in automated workforce management systems.

Methodology:

- **System Architecture Design:** Develop the Eigent desktop platform based on the OWL framework, integrating multi-agent collaboration and task automation modules.
- **Agent Development:** Implement specialized AI agents using machine learning models and reinforcement learning for adaptive task execution.
- **Workflow Configuration:** Design customizable, multi-step workflows that allow agents to coordinate, communicate, and perform tasks autonomously.
- **Parallel Processing & Optimization:** Enable agents to operate concurrently, ensuring efficient task distribution and real-time decision-making.
- **Performance Evaluation:** Continuously monitor agent activities, analyze outcomes, and refine models to enhance productivity, accuracy, and overall system efficiency.
- **Expected Results/Outcomes:**
 - **Enhanced Productivity:** Increased task efficiency through multi-agent parallel execution.
 - **Automation Excellence:** Seamless automation of complex workflows with minimal human input.
 - **Adaptive Learning:** Intelligent agents continuously improving through data-driven feedback.
 - **Data Privacy:** Complete local data handling ensuring user security.
 - **Scalability:** Flexible architecture supporting diverse real-world applications.

Significance/Impact:

Eigent revolutionizes workforce automation by integrating multi-agent intelligence within a desktop environment, enabling seamless collaboration between specialized AI agents. It enhances productivity, reduces human workload, and accelerates decision-making through real-time automation. By prioritizing privacy with local data processing and offering open-source transparency, Eigent empowers organizations to build efficient, scalable, and secure AI-driven workflows for diverse industrial and professional applications.

Key Words:

Artificial Intelligence, Multi-Agent Systems, Workforce Automation, Machine Learning, Desktop Application, Task Optimization, OWL Framework.

Guide

Project Coordinator

HOD



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